

SolutionArchitecture

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TeamID	XXXXXX
ProjectName	EduGenie – Google Gemini Powered Learning Assistant
MaximumMarks	5Marks

SolutionArchitectureDiagram

The **EduGenie – Google Gemini Powered Learning Assistant** follows a layered architecture consisting of the **Presentation Layer**, **Application Layer**, **AI Processing Layer**, and **Data Storage Layer**. Students interact with the web interface to submit learning requests. The FastAPI server processes these requests and communicates with **Google Gemini 1.5 Pro** and **LaMini-Flan-T5** models to generate educational responses. The generated results are stored and displayed through the application interface.

Presentation/ClientLayer

HTML5 • CSS3 • JavaScript •
Responsive Web UI



APIGateway/Server

Python • FastAPI • Uvicorn



Authentication

User Login •
Session Management

CoreLogic

Concept
Explanation •
Question Answering
• Quiz Generation
Text Summarization
• Personalized

ExternalAPIs

Google Gemini 1.5
Pro • LaMini-Flan-T5



Data/StorageLayer

JSON Files • User History • Learning Records • Local Storage

ComponentDescriptionTable

Component Name	Description/RoleinArchitecture	TechnologiesUsed
Presentation Layer	Provides the user interface for students and teachers, including educational features such as concept explanation, Q&A, quiz generation, summarization, and learning recommendations.	HTML5, CSS3, JavaScript
APIGateway	Receives requests from the frontend, validates them, and returns generated responses.	Python, FastAPI, Uvicorn
CoreLogic Service	Processes user requests and manages concept explanation, question answering, quiz generation, summarization, and personalized learning path modules.	Python, FastAPI
Database/ Storage	Stores user information, learning history, generated quizzes, summaries, recommendations, and application data.	JSON Files, Local Storage